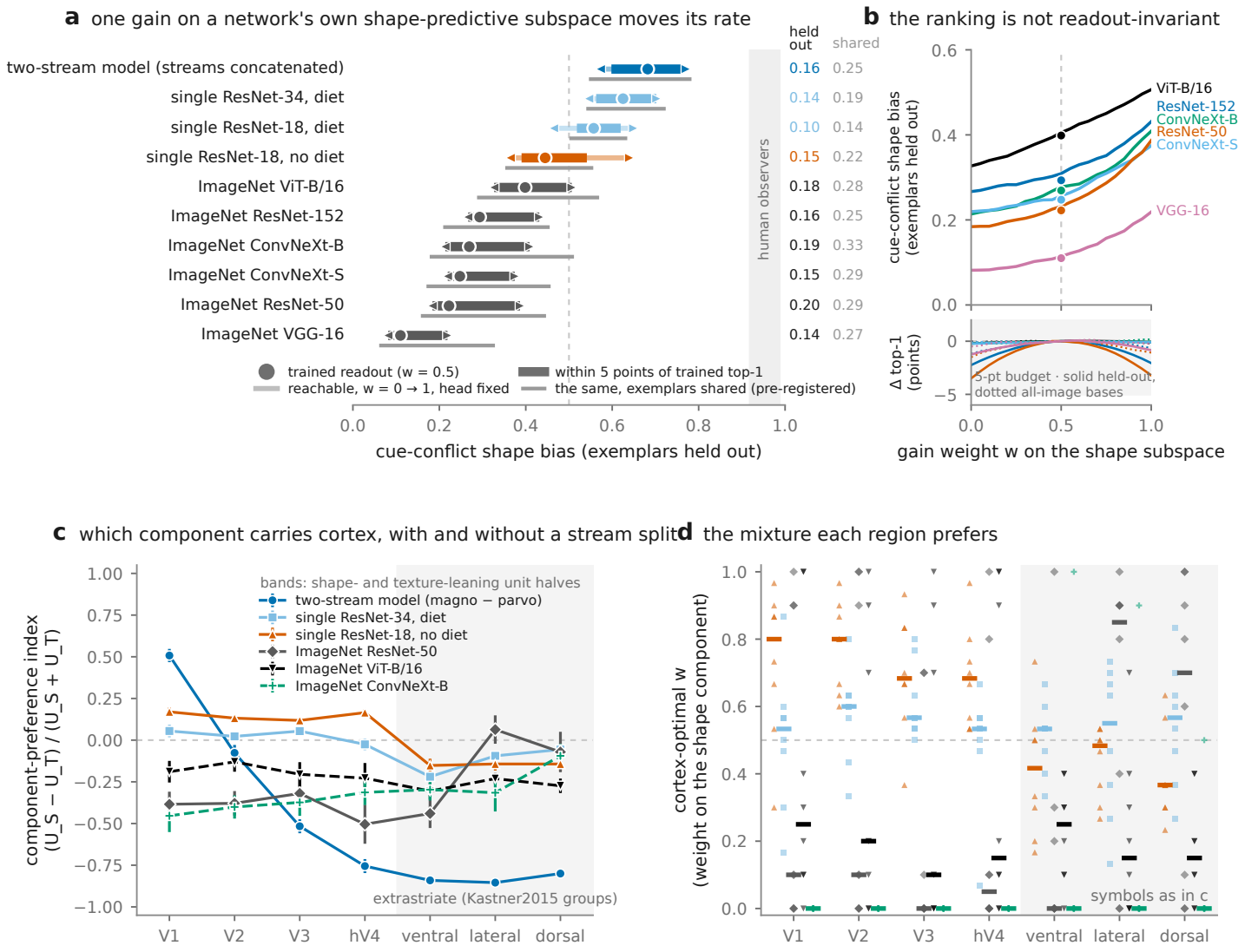
a one gain on a network's own shape-predictive subspace moves its rate
two-stream model (streams concatenated)
single ResNet-34, diet
single ResNet-18, diet
single ResNet-18, no diet
ImageNet ViT-B/16
ImageNet ResNet-152
ImageNet ConvNeXt-B
ImageNet ConvNeXt-S
ImageNet ResNet-50
ImageNet VGG-16
held out
shared
0.16 0.25
0.14 0.19
0.10 0.14
0.15 0.22
0.18 0.28
0.16 0.25
0.19 0.33
0.15 0.29
0.20 0.29
0.14 0.27
human observers
trained readout (w = 0.5)
reachable, w = 0 → 1, head fixed
within 5 points of trained top-1
the same, exemplars shared (pre-registered)
0.0 0.2 0.4 0.6 0.8 1.0
cue-conflict shape bias (exemplars held out)
b the ranking is not readout-invariant
cue-conflict shape bias (exemplars held out)
ViT-B/16
ResNet-152
ConvNeXt-B
ResNet-50
ConvNeXt-S
VGG-16
Δ top-1 (points)
5-pt budget · solid held-out, dotted all-image bases
gain weight w on the shape subspace
c which component carries cortex, with and without a stream split
component-preference index (U_S − U_T) / (U_S + U_T)
bands: shape- and texture-leaning unit halves
two-stream model (magno − parvo)
single ResNet-34, diet
single ResNet-18, no diet
ImageNet ResNet-50
ImageNet ViT-B/16
ImageNet ConvNeXt-B
extrastriate (Kastner2015 groups)
V1 V2 V3 hV4 ventral lateral dorsal
d the mixture each region prefers
cortex-optimal w (weight on the shape component)
symbols as in c
V1 V2 V3 hV4 ventral lateral dorsal